Practical Differentially Private and Byzantine-resilient Federated Learning – Supplementary Material

KEYWORDS

Federated learning, Byzantine Security, Differential privacy, Distributive computation

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A ADDITIONAL DETAILS

A.1 Implementation Detail

Hardware and software setup: The experiments are run on a machine with CentOS Linux release 7.9.2009 (Core). Hardware includes Intel(R) Xeon(R) Gold 6142 CPU @ 2.60GHz and Tesla V100-SXM2-32GB. The code is implemented with PyTorch 1.11.0 and CUDA 10.2. It takes roughly 400 GPU hours to generate all experiment results.

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Model setup: We use CNN network for MNIST and Colorectal and Fully-connected layer for Fashion and USPS, the model size d is 21802, 33736, 25450, 25450 for MNIST, Colorectal, Fashion, and USPS, respectively. Detail setups of network architectures for each dataset are summarised as follows:

1) For MNIST, the network setup is shown in Table 1; For Fashion and USPS, the network setup is shown in Table 2; For Colorectal, We make the CNN for Colorectal have residual connection, due to its complexity, we do not describe it here, the detail can be found in our code.

Layer	Setup
CNN	InputChannel=1, OutputChannel=16, KernelSize=5
ELU	-
GroupNorm	NumGroups=4, NumChannels=16
CNN	InputChannel=16, OutputChannel=16, KernelSize=5
ELU	-
GroupNorm	NumGroups=4, NumChannels=16
CNN	InputChannel=16, OutputChannel=16, KernelSize=5
ELU	-
GroupNorm	NumGroups=4, NumChannels=16
AdaptiveAvgPool	(4,4)
Linear	(256, 32)
ELU	-
Linear	(32, 10)

Table 1: Network setup for MNIST.

Layer	Setup
flatten	-
Linear	(784, 32)
ELU	-
Linear	(32, 10)

Table 2: Network setup for Fasion and USPS.

A.2 List of Notations

We summarise some useful notations we adopt throughout this paper in Table 3

A.3 Byzantine-resilient Aggregation Methods

Giving (gradient) vectors $\{g_i\}_{i=1}^n$, we summarise Krum [2], RFA [7], coordinate-wise median [10] and Trimmed Mean [10] as follows:

- Krum: It returns a vector g_s, which is the closest vector to the mean after excluding $\gamma n + 2$ furthest away points (γ is some constant). Mathematically, let $\mathbb{S} \subseteq \{1, 2, \cdots, n\}$ be an index set with size at least $(n - \gamma n - 2)$. Then,

$$\text{Krum}\,(g_1, \ldots, g_n) = \arg\min_{g_s} \min_{\mathbb{S}} \sum_{j \in \mathbb{S}} \left\| g_s - g_j \right\|^2$$

Notations	Meaning
d	model size
σ	DP noise multiplier
η	learning rate
T	number of total iterations
w^{t-1}	model sent from server to all workers at the beginning of t-th iteration
g_i^t	workers's uplaoding to server from i-th worker at t-th iteration
b_c	local batch size for each worker
D_p	data possessed by the server
f	loss function
G_s^t	set of selected gradient vectors by server at round t

Table 3: Some notations used throughout the paper.

- Robust Federated Averaging (RFA): It just return the geometric median of all vectors:

$$\mathrm{RFA}\left(g_1, \ldots, g_n\right) = \arg\min_g \sum_{i=1}^{n} \|g - g_i\|_2$$

- Coordinate-wise median: Let $v[j]$ refers to the j-th coordinate of a vector v, then the output is just the median of the each coordinate of these vectors:

$$\mathrm{CM}\left(g_1, \ldots, g_n\right)[j] = \mathrm{median}\left(g_1[j], \ldots, g_n[j]\right)$$

- Trimmed Mean: For the j-the coordinate, it computes the average after excluding excluding γn largest and smallest values, *i.e.,*

$$\mathrm{TM}\left(g_1, \ldots, g_n\right)[j] = \frac{1}{n - 2\gamma n} \sum_{k=\gamma n}^{n-\gamma n} g_{\Pi_j(i)}[j],$$

where $g_{\Pi_j(k)}[j]$ represents the k-th largest value among all $\{g_i[j]\}_{s=1}^{n}$.

Algorithm 1 GetNonIID(D, n)

Input: D, the dataset to be distributed to all workers; n, number of workers
1: Partition D (by class) into $G_1, G_2, \cdots, G_H$
2: $D_i \leftarrow [\,]$, $T_i \leftarrow [\,]$, for $i = 1, 2, \cdots, n$
3: **for** $k = 1, 2, \cdots, H$ **do**
4: Generate a list V of uniform R.V.s and normalize V
5: Partition G_k according to V
6: Append parts of G_k to T_i for $i = 1, 2, \cdots, n$
7: **end for**
8: Concatenate all T_i into L
9: $s \leftarrow \left\lceil \frac{|L|}{n} \right\rceil$
10: **for** $i = 1, 2, \cdots, n$ **do**
11: $D_i \leftarrow L[(i-1) \cdot s : i \cdot s]$
12: **end for**
Output: $D_1, D_2, \cdots, D_n$

A.4 Dataset Details and Distribution Setups

Datasets: 1) MNIST [6] is a hand-written digit dataset. It contains 60,000 training examples and 10,000 testing examples, equally distributed among the 10 classes. 2) Colorectal [5], which has 8 classes and 5,000 histological RGB images of size 150×150. The image size is significantly greater than MNIST (which is 28×28). We randomly take 4600 examples for training and take the rest 400 examples for testing. 3) Fashion [9] is an image classification dataset with 10 classes and it has 60,000 training images and 10,000 testing images. 4) USPS [4]. This dataset has 10 classes and 7,291 samples for training and 2,007 for testing.

Generating non-i.i.d. data distribution: The method for generating non-i.i.d. data distributions is summarised in Algorithm 1, and the method for i.i.d. case can be trivially obtained according to line 4. The intuition for why our method can generate non-i.i.d. data distribution is that, according to line 4 in Algorithm 1, the probability that the non-i.i.d. case happens to be i.i.d. is negligible. In practice, applying Algorithm 1 gives us reasonable result which is non-i.i.d. distributed. We provide the non-i.i.d. distribution simulation result for dataset MNIST as an evidence. We can see from Figure 1 that, the data distribution is non-i.i.d.: for a certain class, its sampling probability for each worker is different. Take class 1 for example, its sampling ratio for worker 0 (first sub-figure of first row) is around 0.2, whereas for worker 1 (second sub-figure of first row), it is 0 and for worker 3 (fourth sub-figure of first row), it is almost 0.3. In contrast, for the i.i.d. case, the probability for all class and for all worker should be 0.1.

A.5 Proofs

A.5.1 *Proof of Theorem 1.*

Proof. Our parameter w is a d dimensional vector, considering that our parameter updating rule in the t-th iteration is:

$$w^t = w^{t-1} - \frac{\eta^t}{|B^t|}\left(\sum_{g^t \in B^t} \frac{g^t}{\|g^t\|} + z\right) \tag{1}$$

where B^t is the current batch of per-example gradient (we fix the batch size to be $|B^t| = b_c$), $z \sim \mathcal{N}(0, \sigma^2 I)$ is the DP noise and $g^t = \nabla F(w^{t-1}) + \xi^t$ is the returned result by our stochastic gradient oracle. Considering that we assume F is L-smooth, then we have:

$$\begin{aligned} F(w^t) - F(w^{t-1}) \leq & \left\langle \nabla F(w^{t-1}), w^t - w^{t-1} \right\rangle \\ & + \frac{L}{2}\left\|w^t - w^{t-1}\right\| \end{aligned}$$

which translates to:

$$\begin{aligned} F(w^t) - F(w^{t-1}) \leq & -\frac{\eta^t}{b_c}\left\langle \nabla F(w^{t-1}), \sum_{g^t \in B^t} \frac{g^t}{\|g^t\|} + z \right\rangle \\ & + \frac{L(\eta^t)^2}{2}\left\| \frac{1}{b_c} \sum_{g^t \in B^t} \frac{g^t}{\|g^t\|} + \frac{z}{b_c} \right\| \end{aligned} \tag{2}$$

note that each g^t inside B^t is i.i.d. (because each g^t is calculated by one data example and each data example is sampled from dataset independently with replacement), conditioned on w^{t-1} and take expectation on both side with respect to the randomness of data sampling and DP noise:

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Figure 1: Simulation result for non-i.i.d. distribution for dataset MNIST to 20 workers. 0-9 in the horizontal axis stands for class labels, vertical axis is the ratio of the number of a certain class to the size of the datasets each worker holds. There are 20 subplots, standing for 20 different workers.

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$$
\begin{aligned}
\mathbb{E}(F(w^t)) - F(w^{t-1}) \leq & -\frac{\eta^t}{b_c}\sum_{g^t\in B^t}\mathbb{E}\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle \\
& + \frac{L(\eta^t)^2}{2}\mathbb{E}\left\|\frac{1}{b_c}\sum_{g^t\in B^t}\frac{g^t}{\|g^t\|} + \frac{z}{b_c}\right\| \\
\leq & -\frac{\eta^t}{b_c}\sum_{g^t\in B^t}\mathbb{E}\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle \\
& + \frac{L(\eta^t)^2}{2}(1+\frac{\sigma^2 d}{b_c^2}) \\
= & -\eta^t\mathbb{E}\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle \\
& + \frac{L(\eta^t)^2}{2}\left(1+\frac{\sigma^2 d}{b_c^2}\right)
\end{aligned} \tag{3}
$$

we now deal with the term $\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle$ first, if $\left\|\nabla F(w^{t-1})\right\| > 2\left\|\xi^t\right\|$, we then have:

$$
\begin{aligned}
\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle = & \frac{\left\|\nabla F(w^{t-1})\right\|^2 + \left\langle \nabla F(w^{t-1}), \xi^t\right\rangle}{\left\|\nabla F(w^{t-1}) + \xi^t\right\|} \\
= & \frac{\left\|\nabla F(w^{t-1})\right\|^2}{\left\|\nabla F(w^{t-1}) + \xi^t\right\|} \\
& + \frac{\left\|\nabla F(w^{t-1})\right\|\left\|\xi^t\right\| cos(\theta)}{\left\|\nabla F(w^{t-1}) + \xi^t\right\|} \\
\geq & \frac{\left\|\nabla F(w^{t-1})\right\|^2 - \frac{1}{2}\left\|\nabla F(w^{t-1})\right\|^2}{\left\|\nabla F(w^{t-1}) + \xi^t\right\|} \\
= & \frac{\left\|\nabla F(w^{t-1})\right\|^2}{2\left\|\nabla F(w^{t-1}) + \xi^t\right\|} \\
\geq & \frac{\left\|\nabla F(w^{t-1})\right\|^2}{2\left(\left\|\nabla F(w^{t-1})\right\| + \|\xi^t\|\right)} \\
\geq & \frac{\left\|\nabla F(w^{t-1})\right\|}{3} \\
\geq & \frac{\left\|\nabla F(w^{t-1})\right\|}{3} - \frac{8\left\|\xi^t\right\|}{3}
\end{aligned}
$$

if $\left\|\nabla F(w^{t-1})\right\| \leq 2\left\|\xi^t\right\|$, we then have:

$$\begin{aligned}\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle &\geq -\left\|\nabla F(w^{t-1})\right\| \\ &= \frac{\left\|\nabla F(w^{t-1})\right\|}{3} - \frac{4\left\|\nabla F(w^{t-1})\right\|}{3} \\ &\geq \frac{\left\|\nabla F(w^{t-1})\right\|}{3} - \frac{8\left\|\xi^t\right\|}{3}\end{aligned}$$

Hence we have:

$$\left\langle \nabla F(w^{t-1}), \frac{g^t}{\|g^t\|}\right\rangle \geq \frac{\left\|\nabla F(w^{t-1})\right\|}{3} - \frac{8\left\|\xi^t\right\|}{3} \tag{4}$$

applying Equation 4 to Equation 3 yields:

$$\begin{aligned}\mathbb{E}(F(w^t)) - F(w^{t-1}) \leq &-\eta^t \frac{\mathbb{E}\left\|\nabla F(w^{t-1})\right\|}{3} + \eta^t \frac{8\mathbb{E}\left\|\xi^t\right\|}{3} \\ &+ \frac{L(\eta^t)^2}{2}\left(1 + \frac{\sigma^2 d}{b_c^2}\right)\end{aligned} \tag{5}$$

assume our loss function $F(w^t) > 0$, and the result $g^t = \nabla F(w^{t-1}) + \xi^t$ which is returned by our stochastic gradient oracle has bounded variance, if we set a constant learning rate $\eta^t = \eta$, take iterated expectation, summing up both side from iteration 1 to T, divide both side by $T\eta$, rearrange terms, we have:

$$\begin{aligned}\frac{1}{T}\sum_{t=1}^{t=T}\mathbb{E}\left\|\nabla F(w^t)\right\| \leq &\frac{3F(w^0)}{T\eta} + \frac{3L\eta}{2}\left(1 + \frac{\sigma^2 d}{b_c^2}\right) \\ &+ \frac{8}{T}\sum_{t=1}^{t=T}\mathbb{E}\left\|\xi^t\right\| \\ \leq &\frac{3F(w^0)}{T\eta} + \frac{3L\eta}{2}\left(1 + \frac{\sigma^2 d}{b_c^2}\right) \\ &+ 8\nu\end{aligned} \tag{6}$$

□

A.5.2 Proof of Theorem 3. The following is proof for Theorem 3.

Proof. Algorithm 1 deals with each worker's privacy, we will show that it is (ϵ, δ)-DP for each worker-hold dataset. Note that the l_2-sensitivity of the term $\sum_{j\in[b_c]} \frac{\phi_i[j]}{\|\phi_i[j]\|_2}$ is 2, thus by the same proof in Theorem 1 of [1]. We can see that if $\sigma \geq c_2 \frac{q\sqrt{T\ln\frac{1}{\delta}}}{\epsilon}$, Algorithm 1 guarantee (ϵ, δ)-DP for each worker-hold dataset. □

A.5.3 Proof of Theorem 4. The following is proof for Theorem 4. Before the proof, we first recall some technical lemmas.

Lemma 1 ([8]). *For the Gaussian random variable $x \sim \mathcal{N}(0, \sigma^2 I_d)$, with probability at least $1-\zeta$ for any $\zeta \in (0,1)$ we have*

$$\|\sigma\|_2 \leq \sqrt{2d\log\frac{1}{\zeta}}\sigma$$

Lemma 2 ([3]). *Under Assumption 1 and 2 and assume the parameter space $\Theta \subseteq B(0, r\sqrt{d})$. Then for any $\zeta \in (0,1)$, if $\Delta_1 \leq \frac{\sigma_1^2}{\gamma_1}$ and $\Delta_2 \leq \frac{\sigma_2^2}{\gamma_2}$, then we have the following for all $t \in [T]$:*

$$Pr\{\|g_s^{t-1} - \nabla F(w^{k-1})\| \leq 8\Delta_2\|w^{k-1} - w^*\| + 4\Delta_1\} \geq 1-\zeta,$$

where $\Delta_2 = \sigma_2\sqrt{\frac{2}{|D_p|}}\sqrt{K_1+K_2}$ with $K_1 = d\log\frac{\max\{L,L_1\}}{\sigma_2}$ and $K_2 = \frac{d}{2}\log\frac{|D_p|}{d} + \log\frac{6\sigma^2 r\sqrt{|D_p|}}{\gamma_2\sigma_1\zeta}$, and $\Delta_1 = \sqrt{2}\sigma_1\sqrt{\frac{d\log 6 + \log(3/\delta)}{|D_p|}}$.

Proof. For a fixed iteration t we fist assume the following is true (later we will show it will holds for all $t \in [T]$):

$$\left\|w^{t-1} - w^*\right\| \leq \frac{r}{2}\sqrt{d}.$$

Since we have $\eta_{t-1} = \left\|g_s^{t-1}\right\|_2 \eta_0$, then

$$\begin{aligned}\left\|w^{t-1} - w^*\right\| &= \left\|w^{t-1} - \eta_{t-1}\left(\frac{1}{n}\sum_{g\in G_s^t} g\right) - w^*\right\| \\ &= \|w^{t-1} - \eta_0\left(\frac{\left\|g_s^{t-1}\right\|}{n}\sum_{g\in G_s^t} g\right) \\ &\quad + \eta_0\nabla F\left(w^{t-1}\right) - \eta_0\nabla F\left(w^{t-1}\right) - w^*\| \\ &\leq \underbrace{\left\|w^{t-1} - \eta_0\nabla F\left(w^{t-1}\right) - w^*\right\|}_{A} \\ &\quad + \eta_0\underbrace{\left\|\nabla F\left(w^{t-1}\right) - \frac{\left\|g_s^{t-1}\right\|}{n}\sum_{g\in G_s^t} g\right\|}_{B}.\end{aligned} \tag{7}$$

We first consider term A and we consider the following lemma.

Lemma 3. *[3], Assume Assumption 1 holds. If we set the learning rate satisfies $\alpha \leq \mu/(2L^2)$, then we have the following in any $t \geq 1$:*

$$\left\|w^{t-1} - w^* - \alpha\nabla F\left(w^{t-1}\right)\right\| \leq \sqrt{1 - \frac{\mu^2}{4L^2}}\left\|w^{t-1} - w^*\right\|$$

Proof. Since $\nabla F(w^*) = 0$, we have the following:

$$\begin{aligned}&\left\|w^{t-1} - w^* - \alpha\nabla F\left(w^{t-1}\right)\right\|^2 \\ &= \left\|w^{t-1} - w^* - \alpha\left(\nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right)\right\|^2 \\ &= \left\|w^{t-1} - w^*\right\|^2 + \alpha^2\left\|\nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\|^2 \\ &- 2\alpha\left\langle w^{t-1} - w^*, \nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\rangle\end{aligned} \tag{8}$$

By Assumption 1, we have:

$$\left\|\nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\| \leq L\left\|w^{t-1} - w^*\right\| \tag{9}$$

$$\begin{aligned}&F\left(w^*\right) + \left\langle \nabla F\left(w^*\right), w^{t-1} - w^*\right\rangle \\ &\leq F\left(w^{t-1}\right) - \frac{\mu}{2}\left\|w^{t-1} - w^*\right\|^2\end{aligned} \tag{10}$$

$$F\left(w^{t-1}\right) + \left\langle \nabla F\left(w^{t-1}\right), w^* - w^{t-1}\right\rangle \leq F\left(w^*\right) \tag{11}$$

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Summing up inequalities 10 and 11, we have:

$$\begin{aligned}&\left\langle w^* - w^{t-1}, \nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\rangle \\ &\leq -\frac{\mu}{2}\left\|w^{t-1} - w^*\right\|^2.\end{aligned} \tag{12}$$

Substituting inequalities 9 and 12 into 8, we have:

$$\begin{aligned}&\left\|w^{t-1} - w^* - \alpha\nabla F\left(w^{t-1}\right)\right\|^2 \\ &\qquad\leq \left(1 + \alpha^2 L^2 - \alpha\mu\right)\left\|w^{t-1} - w^*\right\|^2.\end{aligned} \tag{13}$$

By choosing $\alpha \leq \frac{\mu}{2L^2}$, we have:

$$\left\|w^{t-1} - w^* - \alpha\nabla F\left(w^{t-1}\right)\right\|^2 \leq \left(1 - \frac{\mu^2}{4L^2}\right)\left\|w^{t-1} - w^*\right\|^2, \tag{14}$$

which concludes proof for lemma 3. □

For term B, note that,

$$\begin{aligned}&\left\|\nabla F\left(w^{t-1}\right) - \frac{\left\|g_s^{t-1}\right\|}{n}\sum_{g\in G_s^t} g\right\|_2 \\ &\leq \left\|\left\|g_s^{t-1}\right\|\frac{1}{n}\sum_{g\in G_s^t} g - g_s^{t-1}\right\| + \left\|g_s^{t-1} - \nabla F\left(w^{t-1}\right)\right\| \\ &\leq \underbrace{\left\|g_s^{t-1}\right\|\left\|\frac{1}{n}\sum_{g\in G_s^t} g\right\|}_{C} + \left\|g_s^{t-1}\right\| + \left\|g_s^{t-1} - \nabla F\left(w^{t-1}\right)\right\|.\end{aligned} \tag{15}$$

For term C, we can bound it as:

$$\begin{aligned}C &= \left\|g_s^{t-1}\right\|\left\|\frac{1}{n}\sum_{g\in G_s^t} g\right\| \\ &= \left\|g_s^{t-1}\right\|\left\|\frac{1}{n}\sum_{g\in G_s^t} \tilde{g} + z\right\| \\ &\leq \underbrace{\left\|g_s^{t-1}\right\|\left\|\frac{1}{n}\sum_{g\in G_s^t} \tilde{g}\right\|}_{D} + \underbrace{\left\|g_s^{t-1}\right\|\|z\|}_{E},\end{aligned} \tag{16}$$

here $\tilde{g} = \frac{1}{b_c}\sum_{j\in[b_c]}\frac{\phi_i[j]}{\|\phi_i[j]\|_2}$ is just g without noise(refer to line 11) in Algorithm 1 and $z \sim \mathcal{N}(0, \frac{|G_s^t|}{n^2 b_c^2}\sigma^2)$ which is the gaussian noise ensuring DP.

For term D, $D \leq \left\|g_s^{k-1}\right\|$ since $\|\tilde{g}\| \leq 1$ and $|G_s^t| \leq n$; For term E, by Lemma 1 we have with probability at least $1 - \xi$ that,

$$\|z\| \leq \frac{\sqrt{2d|G_s^t|\ln\frac{1}{\xi}}}{nb_c}\sigma \leq \frac{\sqrt{2d\ln\frac{1}{\xi}}}{\sqrt{n}b_c}\sigma.$$

This leads to that, w.p $1 - \xi$,

$$C \leq \left\|g_s^{t-1}\right\|\left(1 + \frac{\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c}\right) \tag{17}$$

Applying this inequality to 15 give us,

$$\begin{aligned}B \leq& 2\left\|g_s^{t-1}\right\| + \left\|g_s^{t-1} - \nabla F\left(w^{t-1}\right)\right\| + \frac{\left\|g_s^{t-1}\right\|\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c} \\ \leq& 3\left\|g_s^{t-1} - \nabla F\left(w^{t-1}\right)\right\| + 2\left\|\nabla F\left(w^{t-1}\right)\right\| \\ &+ \frac{\left\|g_s^{t-1} - \nabla F\left(w^{t-1}\right)\right\|\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c} \\ &+ \frac{\left\|\nabla F\left(w^{t-1}\right)\right\|\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c}\end{aligned} \tag{18}$$

For $\left\|g_s^{t-1} - \nabla F\left(w^{k-1}\right)\right\|$, by Lemma 2 we have, w.p $1 - \xi$,

$$\left\|g_s^{t-1} - \nabla F\left(w^{k-1}\right)\right\| \leq 8\Delta_2\|w^{t-1} - w\| + 4\Delta_1. \tag{19}$$

For $\left\|\nabla F\left(w^{t-1}\right)\right\|$, we have,

$$\begin{aligned}\left\|\nabla F\left(w^{t-1}\right)\right\| &\leq \left\|\nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\| + \left\|\nabla F\left(w^*\right)\right\| \\ &= \left\|\nabla F\left(w^{t-1}\right) - \nabla F\left(w^*\right)\right\| \\ &\leq L\left\|w^{t-1} - w^*\right\|\end{aligned} \tag{20}$$

Applying (19) and (20) to (18), then combine it with Lemma 3 to (7), we have,

$$\begin{aligned}&\left\|w^t - w^*\right\| \leq \sqrt{1 - \frac{u^2}{4L^2}}\left\|w^{t-1} - w^*\right\| \\ &+ \left(24\eta_0\Delta_2 + 2\eta_0 L\right)\left\|w_{t-1} - w^*\right\| \\ &+ \left(8\Delta_2\eta_0 + \eta_0 L\right)\frac{\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c}\left\|w_{k-1} - w^*\right\| \\ &+ \frac{4\eta_0\sqrt{2d\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c}\Delta_1 + 12\eta_0\Delta_1.\end{aligned} \tag{21}$$

Note that for the DP noise multiplier σ (refer to Theorem 3), if we take $\sigma = c_2\frac{b_c\sqrt{T\ln\frac{1}{\delta}}}{|D|\epsilon}$, then we have $\frac{\sqrt{2\ln\frac{1}{\xi}}\sigma}{\sqrt{n}b_c} = \mathcal{O}\left(\frac{\sqrt{T\ln\frac{1}{\delta}\ln\frac{1}{\xi}}}{|D|\sqrt{n}\epsilon}\right)$. Recall our assumption in Theorem 4 that $|D|\sqrt{n} \geq \Omega\left(\frac{\sqrt{T\ln\frac{1}{\delta}\ln\frac{1}{\xi}}\sqrt{d}}{\epsilon}\right)$. Thus, we have $\frac{\sqrt{2T\ln\frac{1}{\xi}}d\sigma}{\sqrt{n}b_c\epsilon} \leq 1$. By taking σ as specified in Theorem

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3, we can update (21) as

$$\begin{aligned}\left\|w^t - w^*\right\| &\leq \left(\sqrt{1-\frac{u^2}{4L^2}} + 32\eta_0\Delta_2 + 3\eta_0 L\right)\left\|w^{t-1} - w^*\right\| \\ &+ \frac{4c_1\sqrt{\ln\frac{1}{\xi}\ln\frac{1}{\delta}T}\eta_0}{\sqrt{n}|D|\epsilon}\Delta_1\sqrt{d} + 12\eta_0\Delta_1 \\ &= (1-\rho)\left\|w^{t-1} - w^*\right\| + O\left(\frac{d\ln\frac{1}{\xi}\sqrt{T\ln\frac{1}{\delta}}\sigma_1}{|D|\sqrt{|D_p|n}\epsilon}\right) \\ &+ O(\eta_0\sigma_1\sqrt{\frac{d}{|D_p|}}),\end{aligned}$$

where $\rho = 1 - \sqrt{1-\frac{u^2}{4L^2}} - 32\eta_0\Delta_2 - 3\eta_0 L$.

Since $\left\|w^{t-1} - w^*\right\| \subset B(0, r\sqrt{d})$, if $|D|\sqrt{|D_p|n}$ is sufficiently large such that $|D|\sqrt{|D_p|n} \geq \Omega\left(\frac{\sqrt{d}\ln\frac{1}{\xi}\sqrt{\ln\frac{1}{\delta}T}}{\epsilon r\rho}\right)$ and $\eta_0\sigma_1/\sqrt{|D_p|} \leq O\left(\frac{r\sqrt{d}}{\rho}\right)$, then $\left\|w^k - w^*\right\| \in B(0, r\sqrt{d})$. Thus, take the union w.r.t all iterations, with probability at least $1 - \xi T$, we have,

$$\begin{aligned}\left\|w^T - w^*\right\| &\leq (1-\rho)^T\left\|w^0 - w^*\right\| \\ &+ O\left(\frac{\eta_0 d\ln\frac{1}{\xi}\sqrt{T\ln\frac{1}{\delta}}\sigma_1}{|D|\sqrt{b_s n}\rho\epsilon} + \frac{\eta_0\Delta_1}{\rho}\right)\end{aligned}$$

By taking $T = O\left(\ln_{1-\rho}\frac{1}{\sqrt{n}|D|\sqrt{|D_p|}}\right) = O\left(\frac{1}{\rho}\ln\left(\sqrt{n}|D|\sqrt{|D_0|}\right)\right)$, we have,

$$\left\|w^T - w^*\right\| \leq \tilde{O}\left(\frac{1}{\rho^2}\frac{d\ln\frac{1}{\xi}\sqrt{\ln\frac{1}{\delta}}\sigma_1}{|D|\sqrt{nb_s}\epsilon} + \frac{1}{\rho}\frac{\sigma_1\sqrt{d\ln\frac{1}{\xi}}}{\sqrt{|D_p|}}\right).$$

□

A.6 More Experimental Results

Experimental results for i.i.d. setting: We present the experimental results on general Byzantine resilience, resilience facing 90% attackers, efficient hyper-parameter tuning in Figure 2, 3, 4 (Gaussian attack), 5, 6, 7 (Optimized Local Model Poisoning attack). As we have observed and analyzed before, similar results can also be observed when our learning protocol is under other attacks.

Experimental results for non-i.i.d. setting: We present all experiments in non-i.i.d. settings as counterparts to previous results. Results are summarise in Figure 8, 9, 10 (Label-flipping attack), 11, 12, 13 (Gaussian attack), 14, 15, 16 (Optimized Local Model Poisoning attack). As we have observed in i.i.d. case, similar results can also be observed when our learning protocol is under various attacks in non-i.i.d. setting.

Resilience against adaptive attacker: We present the remaining results in Figure 17, 18, 19 (i.i.d. setting), 20, 21, 22 (non-i.i.d. setting). All experiments are conducted when there are 60% Byzantine attackers. As we have observed before, our learning protocol is also resilient to adaptive attacker under various attack instantiations.

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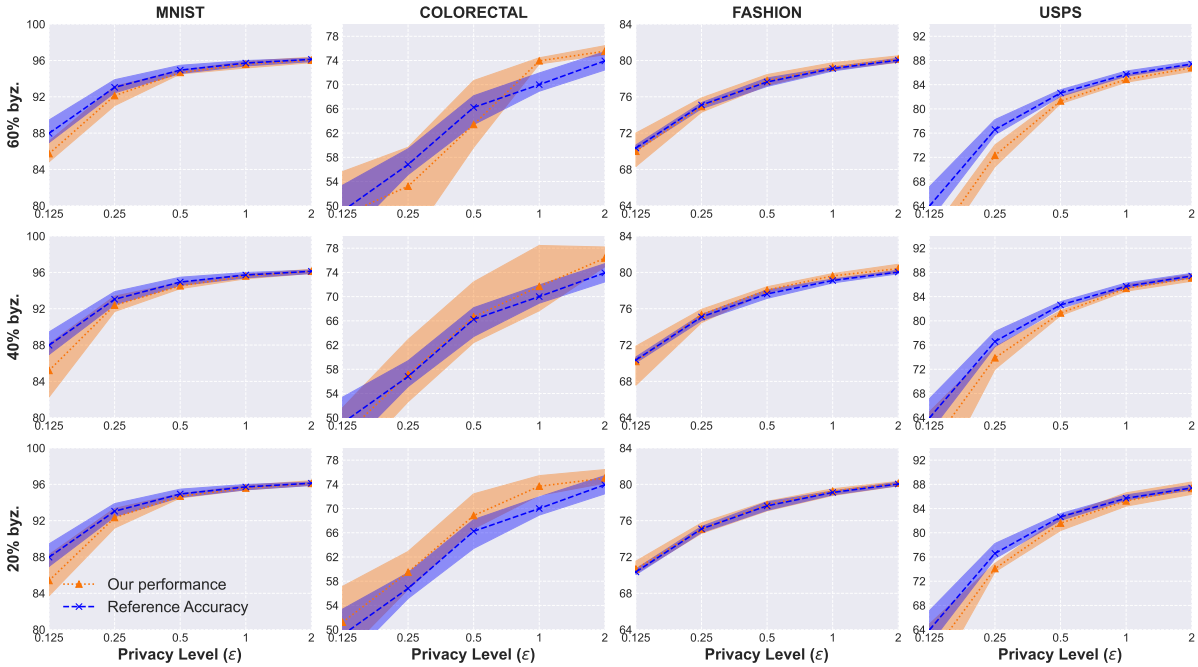

Figure 2: Byzantine-resilient performance (testing accuracy) under Gaussian attack in i.i.d. setting. The experiment is conducted under 3 different attacking levels (20%, 40%, 60% of total worker are Byzantine).

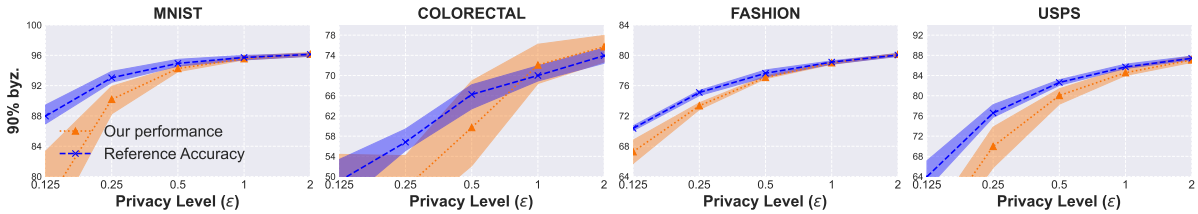

Figure 3: Byzantine-resilient performance (testing accuracy) in i.i.d. setting when 90% workers are Gaussian Byzantine attackers.

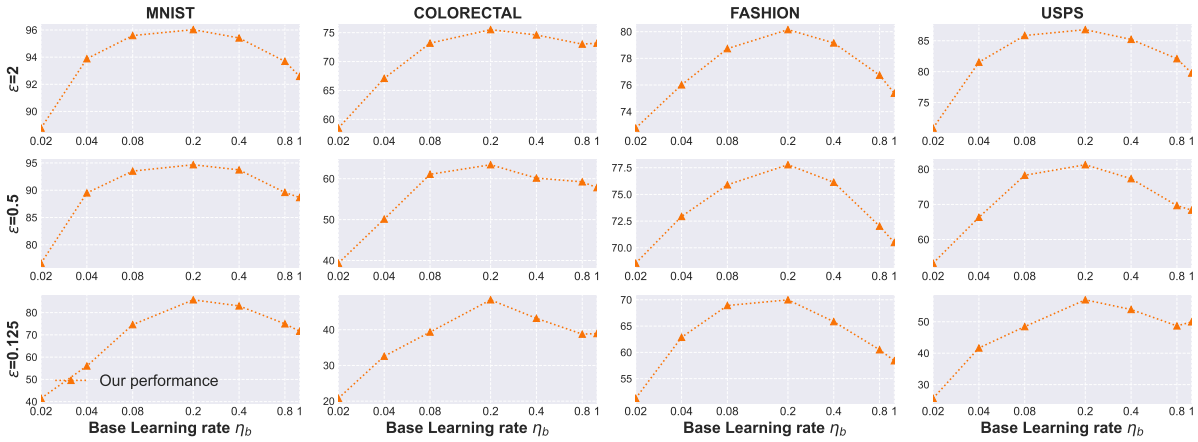

Figure 4: Our hyper-parameter tuning result under i.i.d. setting when facing 60% Gaussian attackers.

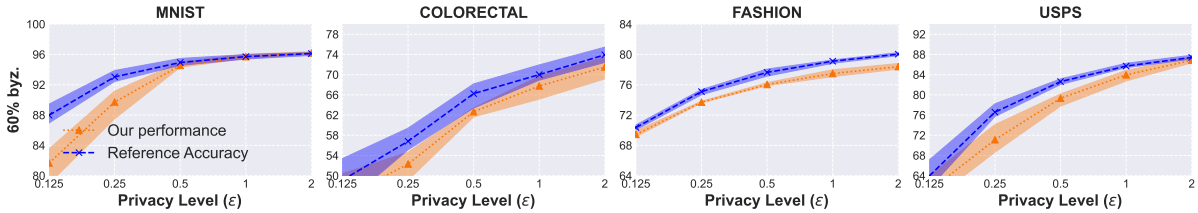

Figure 5: Byzantine-resilient performance (testing accuracy) under Optimized Local Model Poisoning attacks in i.i.d. setting. The experiment is conducted where there are 60% attackers.

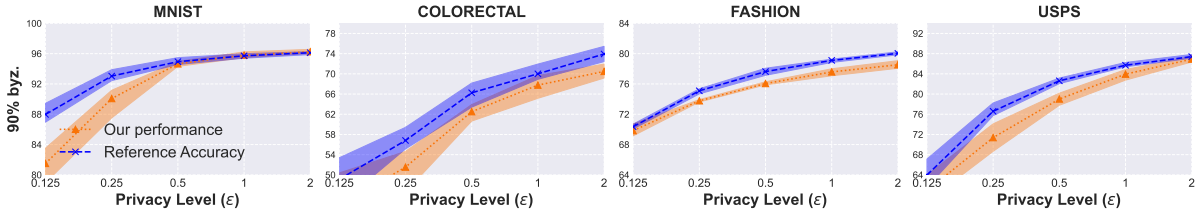

Figure 6: Byzantine-resilient performance (testing accuracy) in i.i.d. setting when 90% workers are Optimized Local Model Poisoning Byzantine attackers.

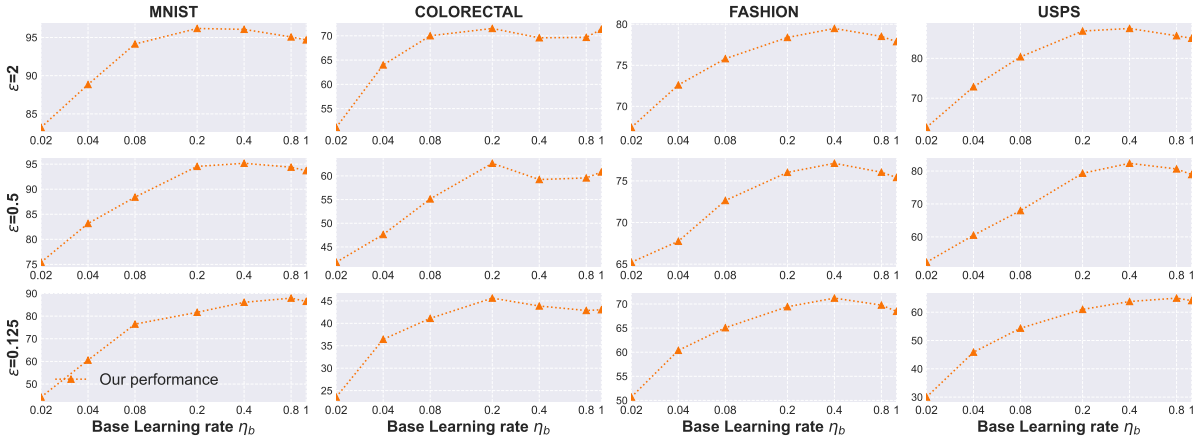

Figure 7: Our hyper-parameter tuning result under i.i.d. setting when facing 60% Optimized Local Model Poisoning attackers.

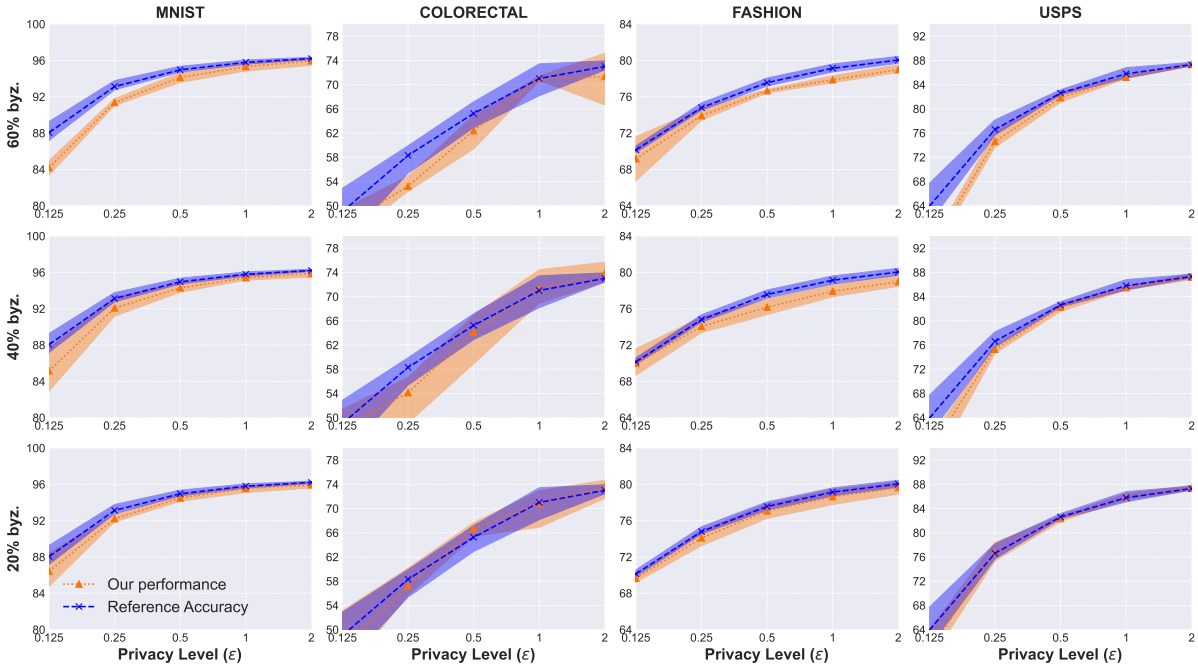

Figure 8: Byzantine-resilient performance (testing accuracy) under Label-flipping attack in non-i.i.d. setting. The experiment is conducted under 3 different attacking levels (20%, 40%, 60% of total worker are Byzantine).

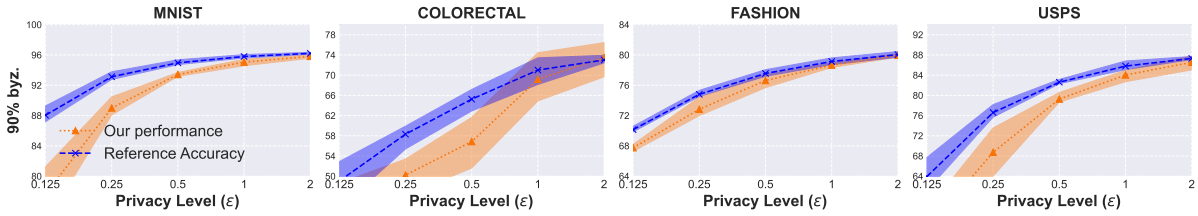

Figure 9: Byzantine-resilient performance (testing accuracy) in non-i.i.d. setting when 90% workers are Label-flipping Byzantine attackers.

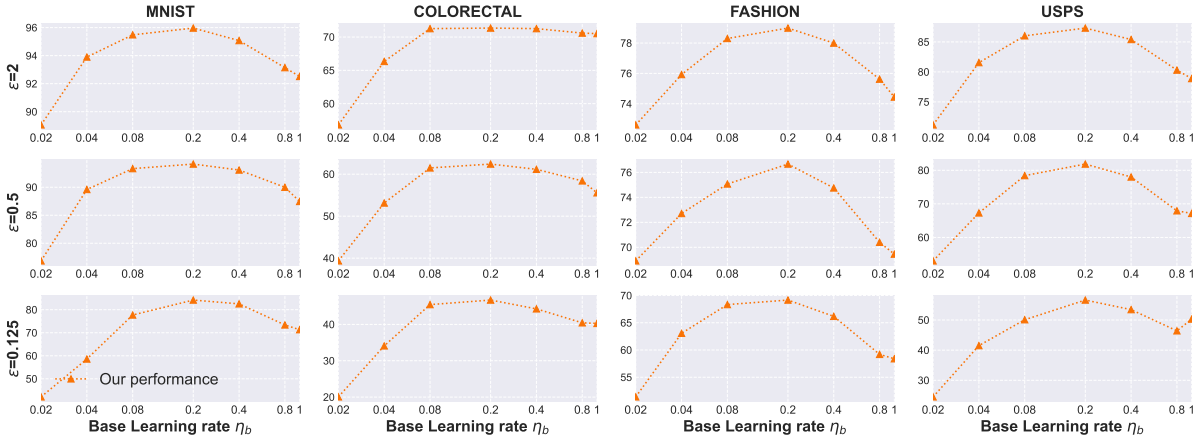

Figure 10: Our hyper-parameter tuning result under non-i.i.d. setting when facing 60% Label-flipping attackers.

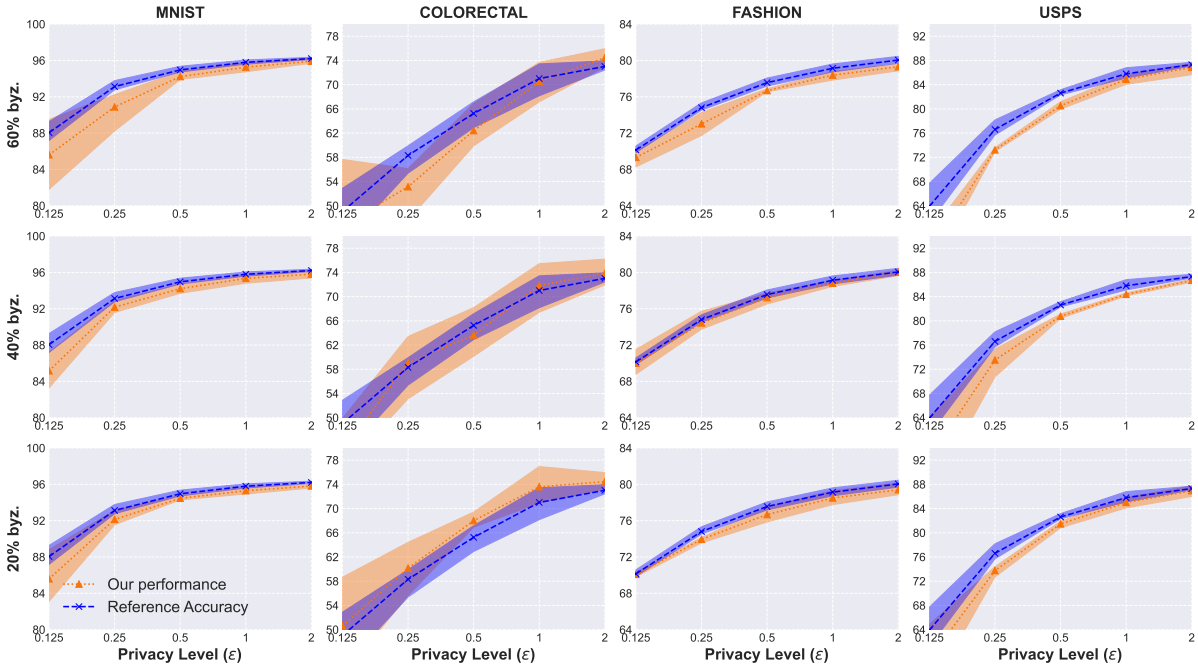

Figure 11: Byzantine-resilient performance (testing accuracy) under Gaussian attack in non-i.i.d. setting. The experiment is conducted under 3 different attacking levels (20%, 40%, 60% of total worker are Byzantine).

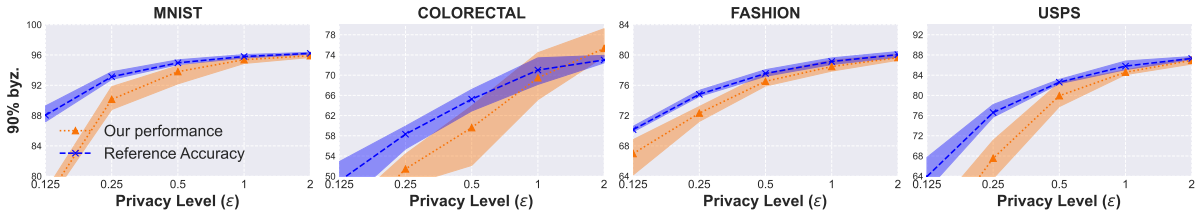

Figure 12: Byzantine-resilient performance (testing accuracy) in non-i.i.d. setting when 90% workers are Gaussian Byzantine attackers.

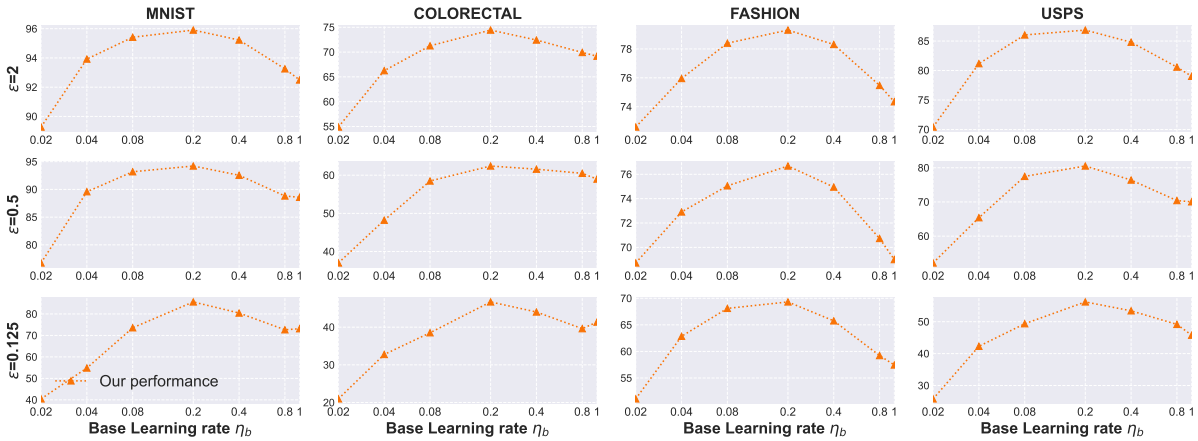

Figure 13: Our hyper-parameter tuning result under non-i.i.d. setting when facing 60% Gaussian attackers.

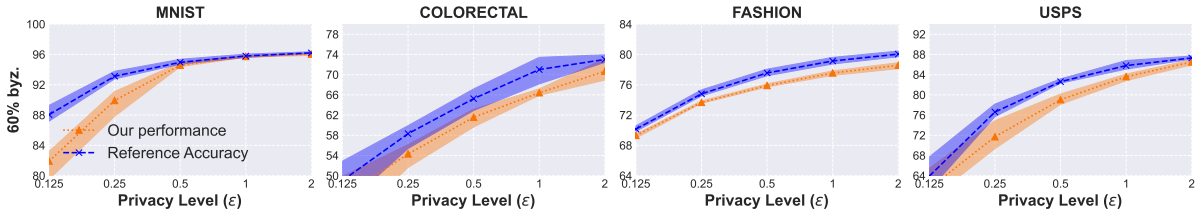

Figure 14: Byzantine-resilient performance (testing accuracy) under Optimized Local Model Poisoning attack in non-i.i.d. setting. The experiment is conducted where there are 60% attackers.

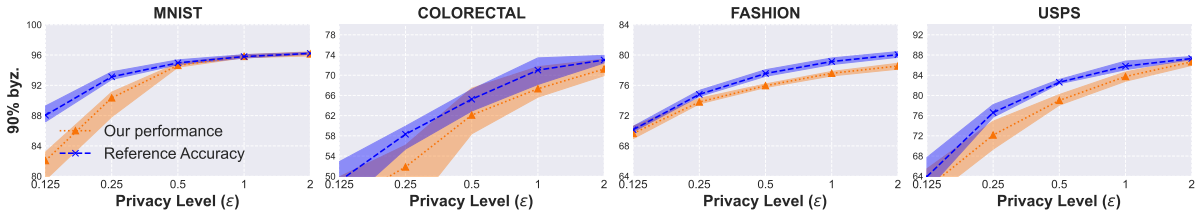

Figure 15: Byzantine-resilient performance (testing accuracy) in non-i.i.d. setting when 90% workers are Optimized Local Model Poisoning Byzantine attackers.

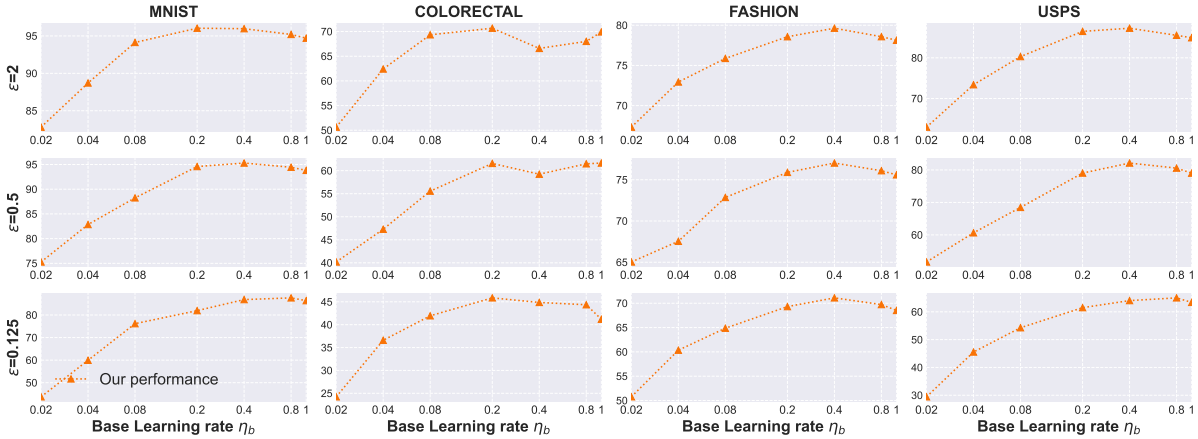

Figure 16: Our hyper-parameter tuning result under non-i.i.d. setting when facing 60% Optimized Local Model Poisoning attackers.

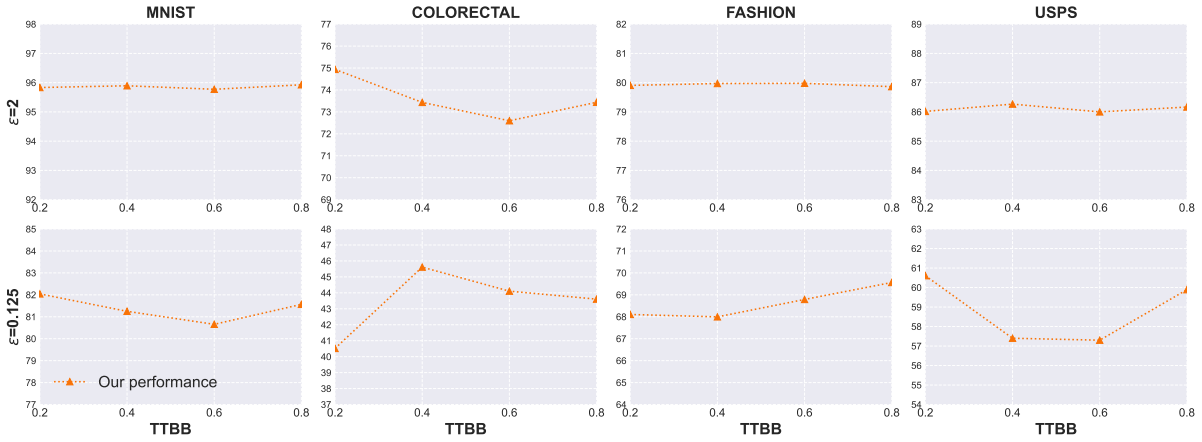

Figure 17: Experiment under Label-flipping attack in i.i.d. setting with different TTBB.

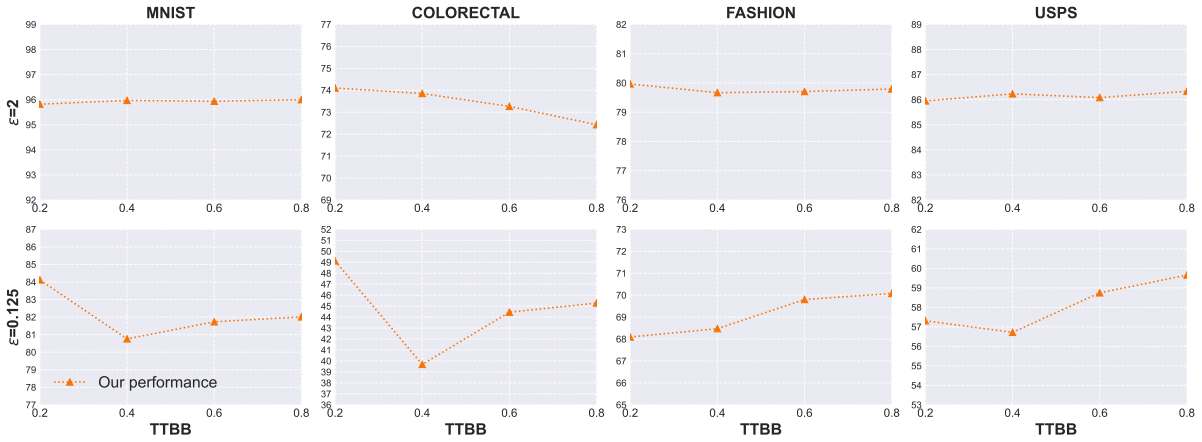

Figure 18: Experiment under Gaussian attack in i.i.d. setting with different TTBB.

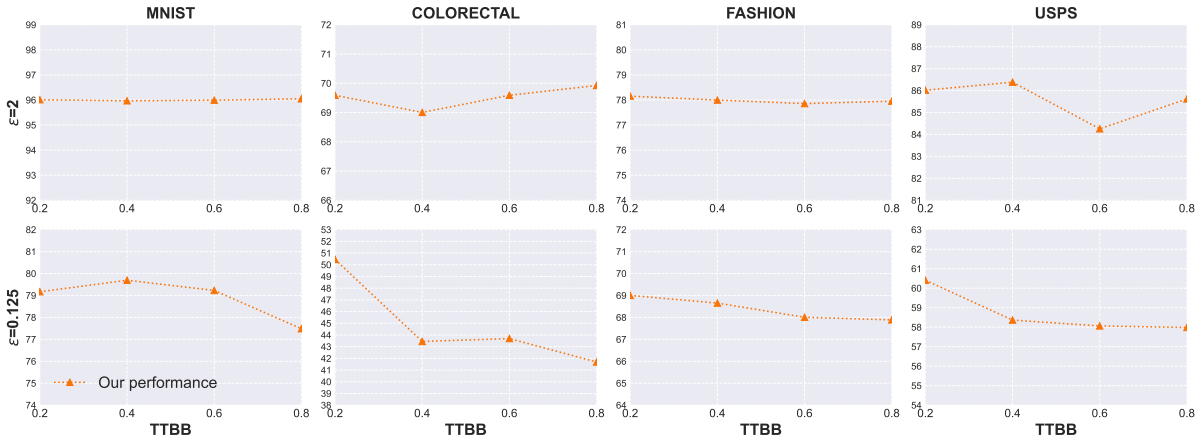

Figure 19: Experiment under Optimized Local Model Poisoning attack in i.i.d. setting with different TTBB.

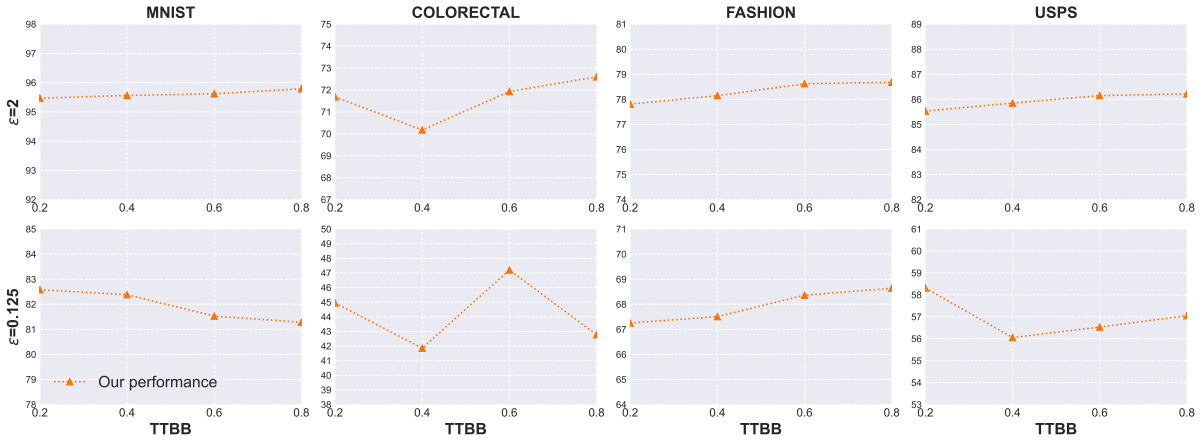

Figure 20: Experiment under Label-flipping attack in non-i.i.d. setting with different TTBB.

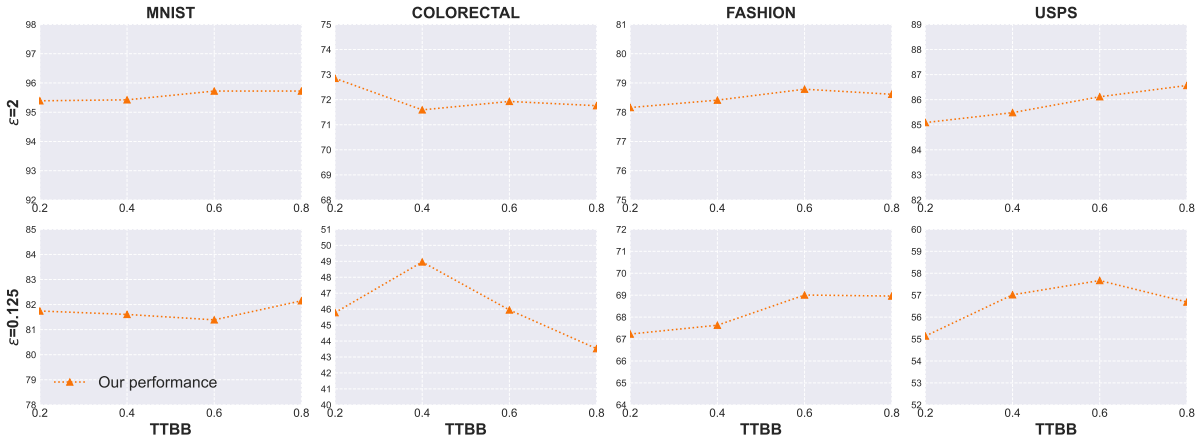

Figure 21: Experiment under Gaussian attack in non-i.i.d. setting with different TTBB.

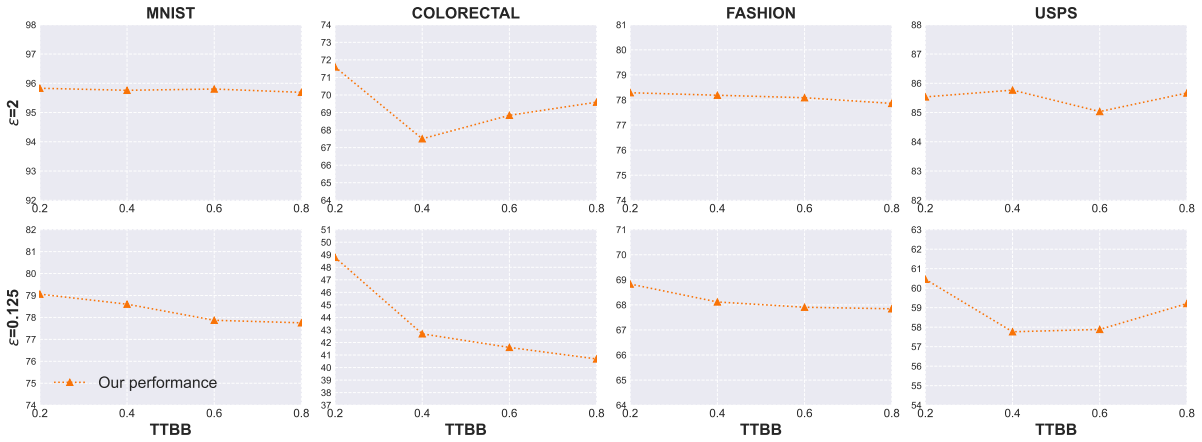

Figure 22: Experiment under Optimized Local Model Poisoning attack in non-i.i.d. setting with different TTBB.